

Rates of Change & Concavity

ANSWER KEY



Average rate of change

- 1 Find the average rate of change of $f(x) = x^2 + 2x$ on the interval $[1, 4]$.

$$\text{AROC} = \frac{f(4) - f(1)}{4 - 1} = \frac{24 - 3}{3} = \frac{21}{3} = 7.$$

- 2 A population grows according to $P(t) = 200(1.05)^t$ (years). Find the average rate of change of the population from $t = 0$ to $t = 10$, to the nearest whole number.

$$P(0) = 200. P(10) = 200(1.05)^{10} \approx 200(1.6289) \approx 325.8. \text{ AROC} = \frac{325.8 - 200}{10} \approx 12.6 \text{ people/year.}$$

Increasing/decreasing behavior from rate of change

- 3 A function f has average rates of change of 2 on $[0, 1]$, 5 on $[1, 2]$, and 9 on $[2, 3]$. Determine whether f appears to be concave up or concave down on $[0, 3]$, and explain.

Since the average rate of change is itself increasing ($2 \rightarrow 5 \rightarrow 9$), f is speeding up as it increases — this is the signature of a concave-up function on $[0, 3]$.

- 4 A different function g has average rates of change of 8 on $[0, 1]$, 5 on $[1, 2]$, and 1 on $[2, 3]$. Classify the concavity of g on $[0, 3]$.

The average rate of change is decreasing ($8 \rightarrow 5 \rightarrow 1$), meaning g 's growth is slowing down — this indicates g is concave down on $[0, 3]$.

Visualizing concavity and inflection

- 5 The graph shows $f(x) = x^3 - 3x$, which changes concavity at $x = 0$.

a Describe the concavity of the graph on $(-2, 0)$. On $(-2, 0)$, the curve bends downward (concave down) — the graph lies below its tangent lines there.

b Describe the concavity of the graph on $(0, 2)$. On $(0, 2)$, the curve bends upward (concave up) — the graph lies above its tangent lines there.

c State what kind of point $x = 0$ represents. $x = 0$ is an inflection point: the location where concavity changes from down to up.

Rate of change of the rate of change

- 6 Using average rates of change over consecutive equal-width intervals as a proxy for how the function's growth rate itself is changing, explain what it means for a function to be "concave up" purely in terms of consecutive AROC values, without any calculus notation.

A function is concave up on an interval when its average rate of change is itself increasing as you move from left to right across consecutive equal-width sub-intervals — the function isn't just increasing, it's increasing at a faster and faster pace (or decreasing at a slower and slower pace).

Applying rate-of-change reasoning to models

- 7 A company's revenue (in thousands) is modeled by $R(t) = t^3 - 6t^2 + 40t$ for $0 \leq t \leq 10$ years. Find the average rate of change on $[0, 2]$ and on $[2, 4]$, then determine whether revenue growth is accelerating or decelerating over $[0, 4]$.

$R(0) = 0$, $R(2) = 8 - 24 + 80 = 64$, $R(4) = 64 - 96 + 160 = 128$. AROC on $[0, 2]$: $\frac{64 - 0}{2} = 32$. AROC on $[2, 4]$: $\frac{128 - 64}{2} = 32$. Since both average rates are equal, revenue growth appears roughly linear (neither clearly accelerating nor decelerating) over this specific interval pair.

- 8 For the same revenue function, find the AROC on $[4, 6]$ and compare it to the AROC on $[2, 4]$ found above, to determine the concavity trend as t increases past 4.

$R(6) = 216 - 216 + 240 = 240$. AROC on $[4, 6]$: $\frac{240 - 128}{2} = 56$. Since $56 > 32$ (the AROC on $[2, 4]$), the rate of change is increasing, indicating the revenue curve is concave up on $[4, 6]$ — growth is accelerating in this later interval.

Estimating rate of change from a table

- 9 A car's odometer reading (miles) is recorded every hour: $t = 0$: 120; $t = 1$: 175; $t = 2$: 238; $t = 3$: 310. Find the AROC over each hour and determine whether the car's speed is increasing, decreasing, or constant.

AROC on $[0, 1]$: $175 - 120 = 55$ mph. AROC on $[1, 2]$: $238 - 175 = 63$ mph. AROC on $[2, 3]$: $310 - 238 = 72$ mph. Since the AROC is increasing each hour ($55 \rightarrow 63 \rightarrow 72$), the car's speed is increasing — consistent with a concave-up distance function.

- 10 Using the same table, find the AROC over the full interval $[0, 3]$ and explain why it differs from the individual hourly rates.

AROC on $[0, 3]$: $\frac{310 - 120}{3} = \frac{190}{3} \approx 63.3$ mph. This is an average over the whole trip, blending the slower early hour with the faster later hour — it necessarily falls between the smallest (55) and largest (72) hourly rates, but doesn't equal any single hourly rate.

Visualizing rate of change on a graph

11 The graph shows $f(x) = \sqrt{x}$, whose average rate of change decreases as x increases.

a Find the AROC of $f(x) = \sqrt{x}$ on $[0, 1]$ and on $[4, 9]$. AROC on $[0, 1]$: $\frac{\sqrt{1} - \sqrt{0}}{1 - 0} = 1$. AROC on $[4, 9]$: $\frac{\sqrt{9} - \sqrt{4}}{9 - 4} = \frac{3 - 2}{5} = 0.2$.

b Explain how the shrinking steepness in the graph relates to your two AROC values. The curve rises steeply near $x = 0$ and flattens out as x increases, matching the drop from an AROC of 1 down to 0.2 — the function is growing, but at an ever-slower pace.

c State whether $f(x) = \sqrt{x}$ is concave up or concave down, based on this pattern. Since the AROC is decreasing as x increases, $f(x) = \sqrt{x}$ is concave down on $(0, \infty)$.

Concavity and real-world interpretation

12 A company reports that its costs are increasing, but at a decreasing rate. Sketch what this implies about the concavity of the cost function, and explain your reasoning without using calculus notation.

"Increasing but at a decreasing rate" means the function is going up (each AROC is positive) but each successive AROC is smaller than the one before — this is the definition of concave down. So the cost function's graph rises but bends downward, flattening out over time.

Multiple choice — easy 15

13 Find the average rate of change of $f(x) = -7x^2$ on $[-4, -1]$.

a. 35 ✓

b. -21

c. 33

d. 37

$$\text{AROC} = \frac{f(-1) - f(-4)}{-1 - (-4)} = \frac{-7 - (-112)}{3} = 35.$$

14 Find the average rate of change of $f(x) = 6x^2 - 4x$ on $[6, -6]$.

- a. -6
- b. -4 ✓
- c. -2
- d. -76

$$\text{AROC} = \frac{f(-6) - f(6)}{-6 - (6)} = \frac{240 - (192)}{-12} = -4.$$

15 Find the average rate of change of $f(x) = 9x^2 + 2x$ on $[5, -3]$.

- a. 18
- b. 20 ✓
- c. -70
- d. 22

$$\text{AROC} = \frac{f(-3) - f(5)}{-3 - (5)} = \frac{75 - (235)}{-8} = 20.$$

16 Find the average rate of change of $f(x) = -5x^2 + 2x$ on $[5, 6]$.

- a. -53 ✓
- b. -55
- c. -3
- d. -51

$$\text{AROC} = \frac{f(6) - f(5)}{6 - (5)} = \frac{-168 - (-115)}{1} = -53.$$

17 Find the average rate of change of $f(x) = -9x^2 + x$ on $[4, 0]$.

- a. -35 ✓
- b. -33
- c. -37
- d. 37

$$\text{AROC} = \frac{f(0) - f(4)}{0 - (4)} = \frac{0 - (-140)}{-4} = -35.$$

18 Find the average rate of change of $f(x) = 7x^2 - 6x$ on $[-1, -2]$.

a. -13

b. -27 ✓

c. -25

d. -29

$$\text{AROC} = \frac{f(-2) - f(-1)}{-2 - (-1)} = \frac{40 - (13)}{-1} = -27.$$

19 Find the average rate of change of $f(x) = 2x^2 - 5x$ on $[6, -3]$.

a. -1

b. -23

c. 1 ✓

d. 3

$$\text{AROC} = \frac{f(-3) - f(6)}{-3 - (6)} = \frac{33 - (42)}{-9} = 1.$$

20 Find the average rate of change of $f(x) = 3x^2 - 4x$ on $[3, 5]$.

a. 22

b. 2

c. 20 ✓

d. 18

$$\text{AROC} = \frac{f(5) - f(3)}{5 - (3)} = \frac{55 - (15)}{2} = 20.$$

21 Find the average rate of change of $f(x) = -4x^2 + 9x$ on $[1, -2]$.

a. 15

b. 13 ✓

c. 21

d. 11

$$\text{AROC} = \frac{f(-2) - f(1)}{-2 - (1)} = \frac{-34 - (5)}{-3} = 13.$$

22 Find the average rate of change of $f(x) = 6x^2$ on $[3, -3]$.

a. 0 ✓

b. -36

c. 2

d. -2

$$\text{AROC} = \frac{f(-3) - f(3)}{-3 - (3)} = \frac{54 - (54)}{-6} = 0.$$

23 Find the average rate of change of $f(x) = -3x^2$ on $[1, -4]$.

a. 11

b. 9 ✓

c. 15

d. 7

$$\text{AROC} = \frac{f(-4) - f(1)}{-4 - (1)} = \frac{-48 - (-3)}{-5} = 9.$$

24 Find the average rate of change of $f(x) = -9x^2 - 9x$ on $[-4, 0]$.

a. -45

b. 25

c. 29

d. 27 ✓

$$\text{AROC} = \frac{f(0) - f(-4)}{0 - (-4)} = \frac{0 - (-108)}{4} = 27.$$

25 Find the average rate of change of $f(x) = 9x^2 + 8x$ on $[4, -3]$.

a. -55

b. 19

c. 17 ✓

d. 15

$$\text{AROC} = \frac{f(-3) - f(4)}{-3 - (4)} = \frac{57 - (176)}{-7} = 17.$$

26 Find the average rate of change of $f(x) = x^2 + 4x$ on $[5, 4]$.

a. 11

b. 15

c. 13 ✓

d. 3

$$\text{AROC} = \frac{f(4) - f(5)}{4 - (5)} = \frac{32 - (45)}{-1} = 13.$$

27 Find the average rate of change of $f(x) = -6x^2 + x$ on $[-3, -2]$.

a. 29

b. 33

c. -5

d. 31 ✓

$$\text{AROC} = \frac{f(-2) - f(-3)}{-2 - (-3)} = \frac{-26 - (-57)}{1} = 31.$$

Multiple choice — medium 15

28 Find the average rate of change of $f(x) = -2x^3$ on $[2, -5]$.

a. -38 ✓

b. -42

c. -18

d. -34

$$\text{AROC} = \frac{f(-5) - f(2)}{-5 - (2)} = \frac{250 - (-16)}{-7} = -38.$$

29 Find the average rate of change of $f(x) = -2x^3$ on $[-1, 5]$.

a. -42 ✓

b. -46

c. -32

d. -38

$$\text{AROC} = \frac{f(5) - f(-1)}{5 - (-1)} = \frac{-250 - (2)}{6} = -42.$$

30 Find the average rate of change of $f(x) = 3x^3$ on $[2, -1]$.

a. 15

b. 3

c. 11

d. 9 ✓

$$\text{AROC} = \frac{f(-1) - f(2)}{-1 - (2)} = \frac{-3 - (24)}{-3} = 9.$$

31 Find the average rate of change of $f(x) = -x^3$ on $[0, 5]$.

a. -21

b. -25 ✓

c. -27

d. -23

$$\text{AROC} = \frac{f(5) - f(0)}{5 - (0)} = \frac{-125 - (0)}{5} = -25.$$

32 Find the average rate of change of $f(x) = -3x^3$ on $[4, 5]$.

a. -183 ✓

b. -189

c. -243

d. -177

$$\text{AROC} = \frac{f(5) - f(4)}{5 - (4)} = \frac{-375 - (-192)}{1} = -183.$$

33 Find the average rate of change of $f(x) = 3x^3$ on $[-2, 3]$.

a. 27

b. 15

c. 21 ✓

d. 3

$$\text{AROC} = \frac{f(3) - f(-2)}{3 - (-2)} = \frac{81 - (-24)}{5} = 21.$$

34 Find the average rate of change of $f(x) = 4x^3$ on $[0, -3]$.

a. 40

b. 36 ✓

c. 44

d. 28

$$\text{AROC} = \frac{f(-3) - f(0)}{-3 - (0)} = \frac{-108 - (0)}{-3} = 36.$$

35 Find the average rate of change of $f(x) = 2x^3$ on $[-4, 1]$.

- a. 30
- b. 22
- c. 26 ✓
- d. 18

$$\text{AROC} = \frac{f(1) - f(-4)}{1 - (-4)} = \frac{2 - (-128)}{5} = 26.$$

36 Find the average rate of change of $f(x) = 4x^3$ on $[2, -4]$.

- a. 16
- b. 48 ✓
- c. 40
- d. 56

$$\text{AROC} = \frac{f(-4) - f(2)}{-4 - (2)} = \frac{-256 - (32)}{-6} = 48.$$

37 Find the average rate of change of $f(x) = -4x^3$ on $[0, 2]$.

- a. -24
- b. -16 ✓
- c. -8
- d. -12

$$\text{AROC} = \frac{f(2) - f(0)}{2 - (0)} = \frac{-32 - (0)}{2} = -16.$$

38 Find the average rate of change of $f(x) = -5x^3$ on $[-2, 3]$.

- a. -25
- b. -5
- c. -35 ✓
- d. -45

$$\text{AROC} = \frac{f(3) - f(-2)}{3 - (-2)} = \frac{-135 - (40)}{5} = -35.$$

39 Find the average rate of change of $f(x) = -5x^3$ on $[1, -2]$.

a. -15 ✓

b. -19

c. -5

d. -25

$$\text{AROC} = \frac{f(-2) - f(1)}{-2 - (1)} = \frac{40 - (-5)}{-3} = -15.$$

40 Find the average rate of change of $f(x) = -5x^3$ on $[0, -3]$.

a. -47

b. -35

c. -55

d. -45 ✓

$$\text{AROC} = \frac{f(-3) - f(0)}{-3 - (0)} = \frac{135 - (0)}{-3} = -45.$$

41 Find the average rate of change of $f(x) = -x^3$ on $[2, 3]$.

a. -19 ✓

b. -25

c. -17

d. -21

$$\text{AROC} = \frac{f(3) - f(2)}{3 - (2)} = \frac{-27 - (-8)}{1} = -19.$$

42 Find the average rate of change of $f(x) = 6x^3$ on $[3, -5]$.

- a. 24
- b. 102
- c. 126
- d. 114 ✓

$$\text{AROC} = \frac{f(-5) - f(3)}{-5 - (3)} = \frac{-750 - (162)}{-8} = 114.$$

Multiple choice — hard 15

43 A car's odometer reading (miles) is recorded every hour: $t = 0$: 131; $t = 1$: 150; $t = 2$: 178; $t = 3$: 215. Is the car's speed increasing, decreasing, or constant?

- a. Decreasing
- b. Constant
- c. Cannot be determined from this data
- d. Increasing ✓

AROC on $[0, 1]$: 19 mph. AROC on $[1, 2]$: 28 mph. AROC on $[2, 3]$: 37 mph. The hourly rates are increasing, so the car's speed is Increasing.

44 A car's odometer reading (miles) is recorded every hour: $t = 0$: 75; $t = 1$: 111; $t = 2$: 138; $t = 3$: 156. Is the car's speed increasing, decreasing, or constant?

- a. Decreasing ✓
- b. Increasing
- c. Constant
- d. Cannot be determined from this data

AROC on $[0, 1]$: 36 mph. AROC on $[1, 2]$: 27 mph. AROC on $[2, 3]$: 18 mph. The hourly rates are decreasing, so the car's speed is Decreasing.

- 45 A car's odometer reading (miles) is recorded every hour: $t = 0$: 62; $t = 1$: 81; $t = 2$: 104; $t = 3$: 136. Is the car's speed increasing, decreasing, or constant?
- a. Constant
 - b. Cannot be determined from this data
 - c. Decreasing
 - d. **Increasing ✓**

AROC on $[0, 1]$: 19 mph. AROC on $[1, 2]$: 23 mph. AROC on $[2, 3]$: 32 mph. The hourly rates are increasing, so the car's speed is Increasing.

- 46 A car's odometer reading (miles) is recorded every hour: $t = 0$: 133; $t = 1$: 148; $t = 2$: 168; $t = 3$: 194. Is the car's speed increasing, decreasing, or constant?
- a. Cannot be determined from this data
 - b. Decreasing
 - c. **Increasing ✓**
 - d. Constant

AROC on $[0, 1]$: 15 mph. AROC on $[1, 2]$: 20 mph. AROC on $[2, 3]$: 26 mph. The hourly rates are increasing, so the car's speed is Increasing.

- 47 A car's odometer reading (miles) is recorded every hour: $t = 0$: 151; $t = 1$: 174; $t = 2$: 197; $t = 3$: 220. Is the car's speed increasing, decreasing, or constant?
- a. Increasing
 - b. **Constant ✓**
 - c. Decreasing
 - d. Cannot be determined from this data

AROC on $[0, 1]$: 23 mph. AROC on $[1, 2]$: 23 mph. AROC on $[2, 3]$: 23 mph. The hourly rates are constant, so the car's speed is Constant.

- 48 A function h has average rates of change of 4 on $[0, 1]$, 2 on $[1, 2]$, and 1 on $[2, 3]$. Classify the concavity of h on $[0, 3]$.
- a. Concave up
 - b. Cannot be determined from this data
 - c. Linear (neither concave up nor down)
 - d. **Concave down ✓**

The average rate of change is decreasing ($4 \rightarrow 2 \rightarrow 1$), so h is concave down on $[0, 3]$.

- 49 A function h has average rates of change of 4 on $[0, 1]$, 7 on $[1, 2]$, and 12 on $[2, 3]$. Classify the concavity of h on $[0, 3]$.
- a. Linear (neither concave up nor down)
 - b. Cannot be determined from this data
 - c. Concave down
 - d. **Concave up ✓**

The average rate of change is increasing ($4 \rightarrow 7 \rightarrow 12$), so h is concave up on $[0, 3]$.

- 50 A function h has average rates of change of 5 on $[0, 1]$, 7 on $[1, 2]$, and 11 on $[2, 3]$. Classify the concavity of h on $[0, 3]$.
- a. Linear (neither concave up nor down)
 - b. Concave down
 - c. **Concave up ✓**
 - d. Cannot be determined from this data

The average rate of change is increasing ($5 \rightarrow 7 \rightarrow 11$), so h is concave up on $[0, 3]$.

- 51 A function h has average rates of change of 9 on $[0, 1]$, 8 on $[1, 2]$, and 6 on $[2, 3]$. Classify the concavity of h on $[0, 3]$.

- a. Cannot be determined from this data
- b. Linear (neither concave up nor down)
- c. Concave down ✓
- d. Concave up

The average rate of change is decreasing ($9 \rightarrow 8 \rightarrow 6$), so h is concave down on $[0, 3]$.

- 52 A function h has average rates of change of 5 on $[0, 1]$, 0 on $[1, 2]$, and -1 on $[2, 3]$. Classify the concavity of h on $[0, 3]$.

- a. Concave down ✓
- b. Linear (neither concave up nor down)
- c. Cannot be determined from this data
- d. Concave up

The average rate of change is decreasing ($5 \rightarrow 0 \rightarrow -1$), so h is concave down on $[0, 3]$.

- 53 Find the average rate of change of $f(x) = 6\sqrt{x}$ on $[4, 25]$.

- a. $\frac{13}{7}$
- b. $\frac{6}{7}$ ✓
- c. 2
- d. $\frac{1}{7}$

$$\text{AROC} = \frac{6\sqrt{25} - 6\sqrt{4}}{25 - 4} = \frac{30 - 12}{21} = \frac{6}{7}.$$

54 Find the average rate of change of $f(x) = 9\sqrt{x}$ on $[1, 9]$.

a. $\frac{13}{4}$

b. $\frac{9}{4}$ ✓

c. $\frac{9}{2}$

d. $\frac{1}{4}$

$$\text{AROC} = \frac{9\sqrt{9} - 9\sqrt{1}}{9 - 1} = \frac{27 - 9}{8} = \frac{9}{4}.$$

55 Find the average rate of change of $f(x) = 7\sqrt{x}$ on $[9, 49]$.

a. $\frac{7}{10}$ ✓

b. $\frac{7}{4}$

c. $\frac{1}{10}$

d. $\frac{17}{10}$

$$\text{AROC} = \frac{7\sqrt{49} - 7\sqrt{9}}{49 - 9} = \frac{49 - 21}{40} = \frac{7}{10}.$$

56 Find the average rate of change of $f(x) = 5\sqrt{x}$ on $[16, 49]$.

a. $\frac{5}{11}$ ✓

b. $\frac{1}{11}$

c. $\frac{16}{11}$

d. $\frac{5}{3}$

$$\text{AROC} = \frac{5\sqrt{49} - 5\sqrt{16}}{49 - 16} = \frac{35 - 20}{33} = \frac{5}{11}.$$

57 Find the average rate of change of $f(x) = 4\sqrt{x}$ on $[4, 36]$.

a. 1

b. $\frac{1}{8}$

c. $\frac{1}{2}$ ✓

d. $\frac{3}{2}$

$$\text{AROC} = \frac{4\sqrt{36} - 4\sqrt{4}}{36 - 4} = \frac{24 - 8}{32} = \frac{1}{2}.$$